

Seminar 1 - Quiz

✓✓ (1). Equipollence relation:

A. equivalence relation.

✓✓ (2) $\vec{OA'} + \vec{OB'} + \vec{OC'} = c \cdot (\vec{OA} + \vec{OB} + \vec{OC})$.

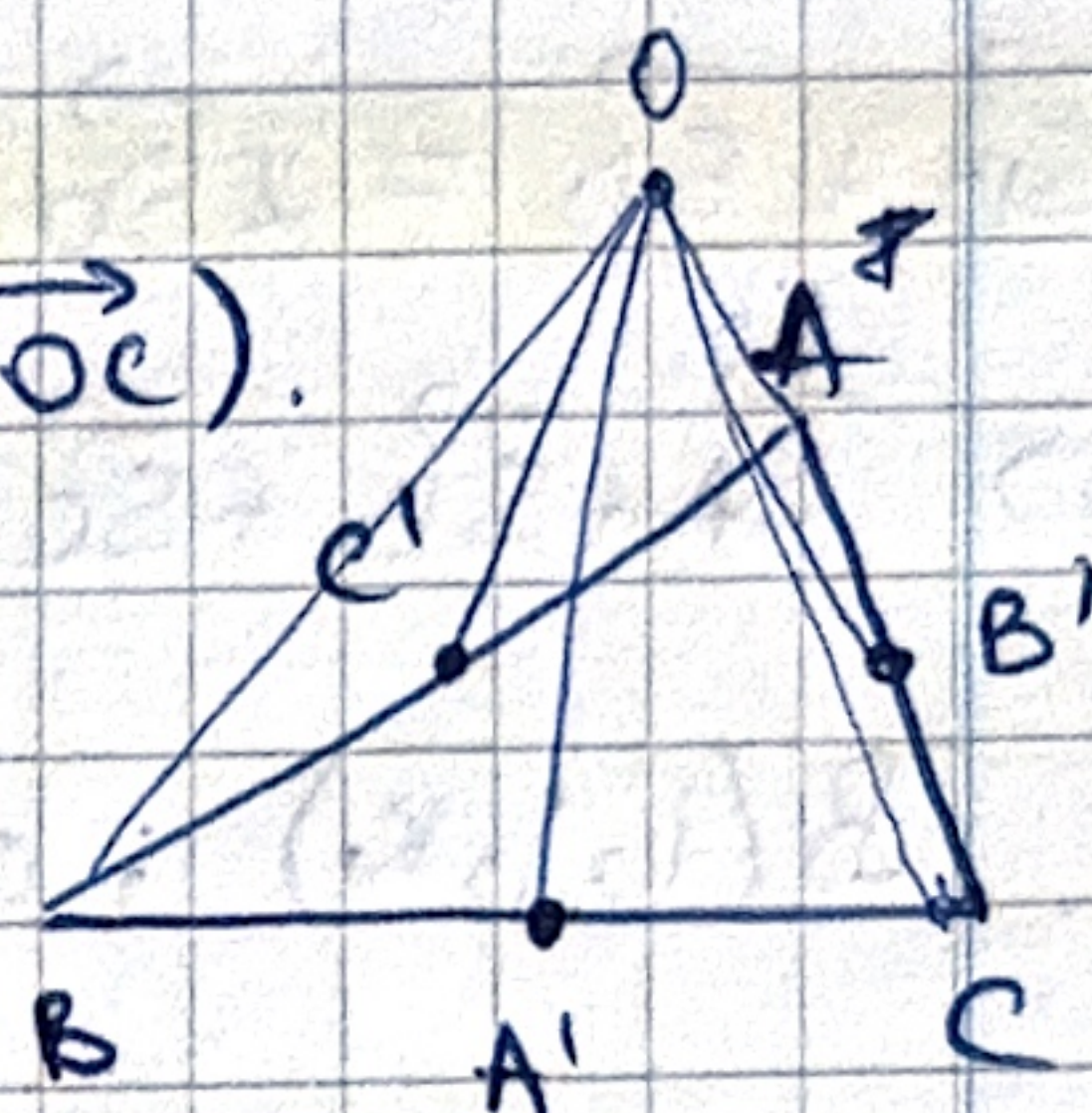
$$\vec{OA'} = \frac{1}{2} (\vec{OB} + \vec{OC})$$

$$\vec{OB'} = \frac{1}{2} (\vec{OA} + \vec{OC})$$

$$\vec{OC'} = \frac{1}{2} (\vec{OA} + \vec{OB})$$

————— (+)

$$\vec{OA'} + \vec{OB'} + \vec{OC'} = \frac{1}{2} (2\vec{OA} + 2\vec{OB} + 2\vec{OC}) \Rightarrow \boxed{c=1.}$$



✓✓ (3). ABCD tetrahedron.

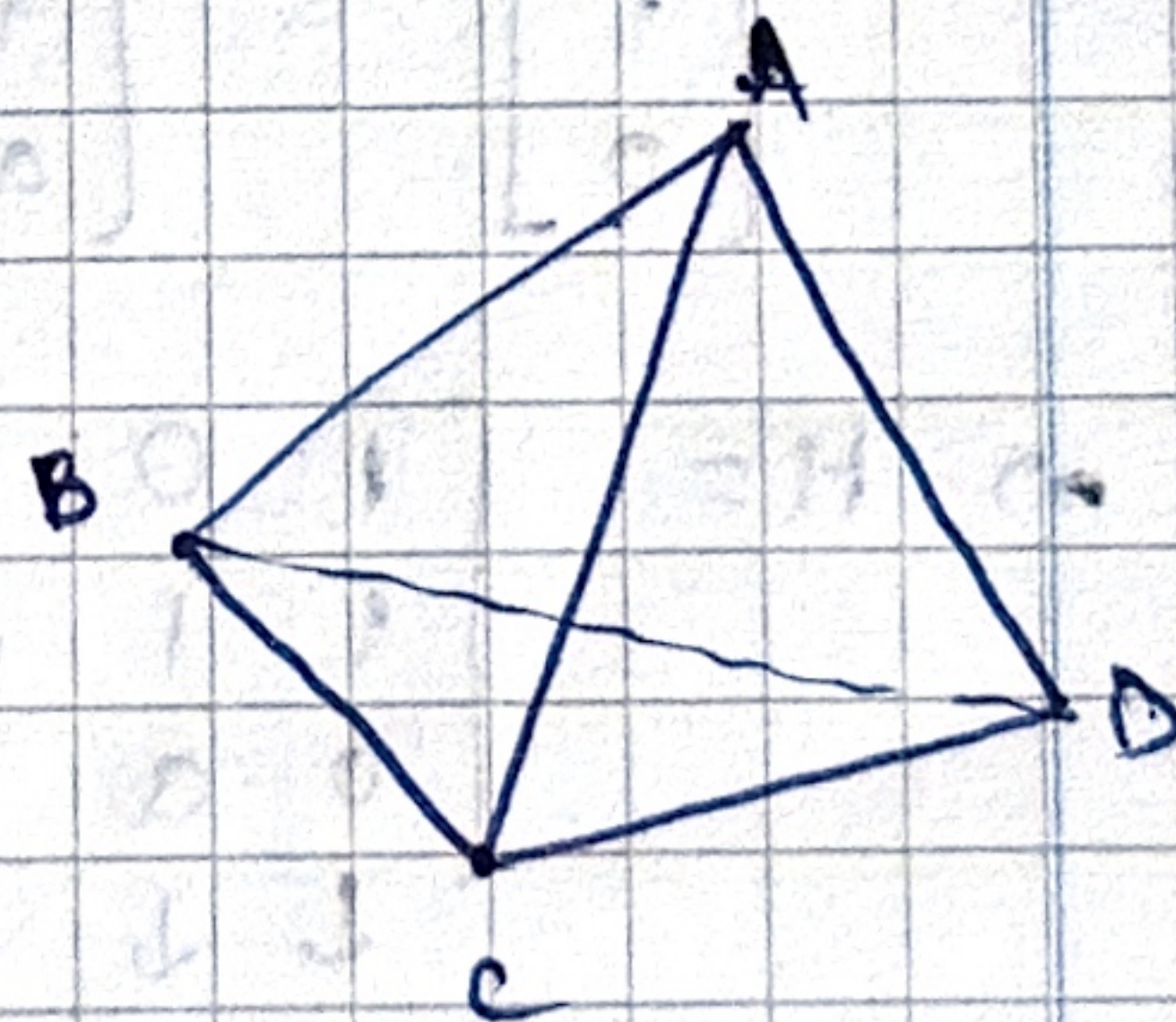
A. $\vec{AB} + \vec{BC} + \vec{CD} = \vec{AD}$

$$\vec{AC} + \vec{CD} = \vec{AD}$$

B. $\vec{AD} + \vec{DB} + \vec{BC} = \vec{AC}$

C. —

D. $\vec{AB} + \vec{BC} + \vec{CD} + \vec{DA} = \vec{AA} = \vec{0}$.



✓✓ (4). ABCD tetrahedron.

$$\vec{AD} + \vec{BC} = \vec{BD} + \vec{u}$$

$$(\Rightarrow) \vec{AD} - \vec{BD} = \vec{u} - \vec{BC} \quad (\Rightarrow) \vec{AD} + \vec{DB} = \vec{u} + \vec{CB}$$

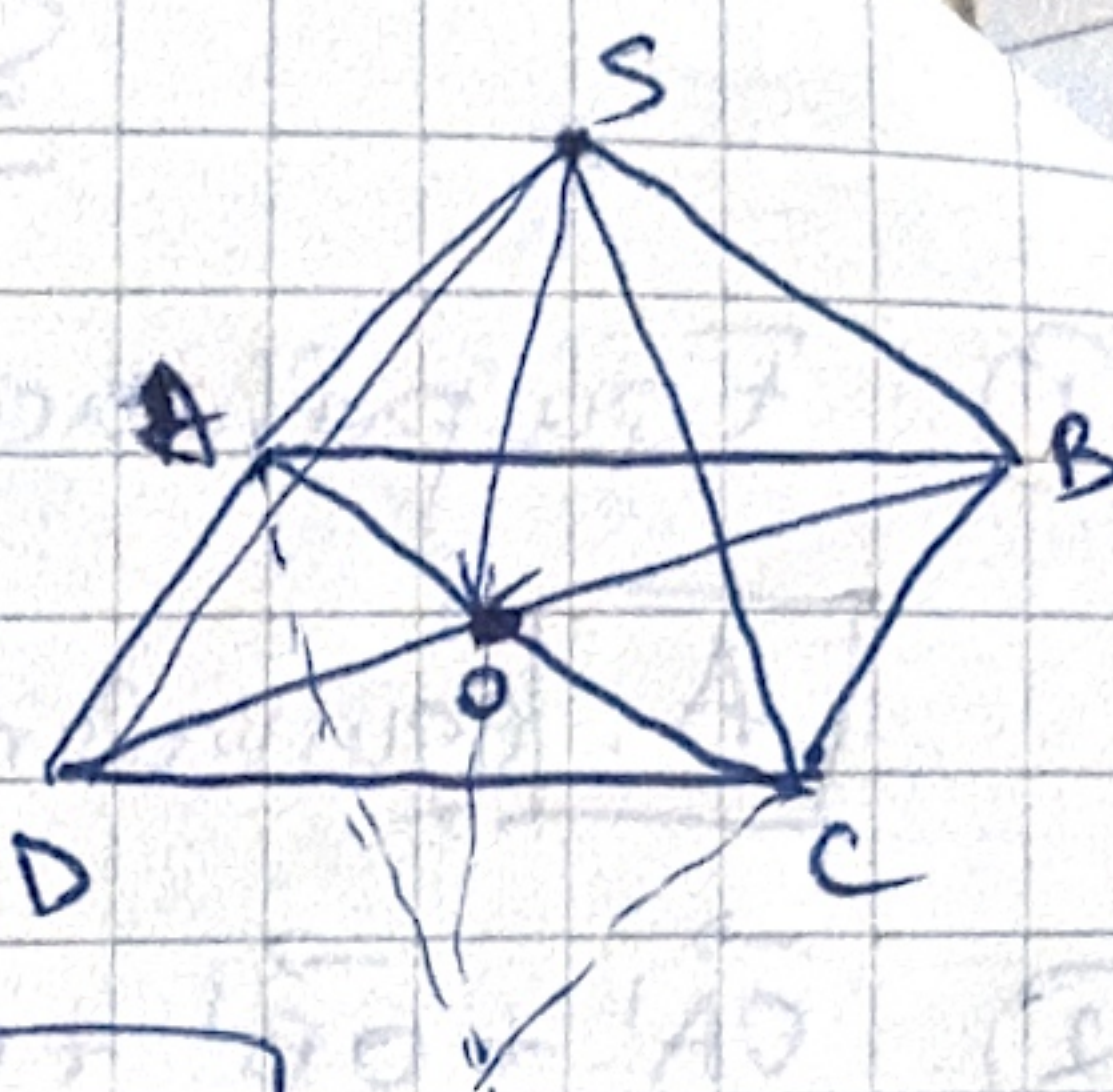
$$(\Rightarrow) \vec{AB} = \vec{u} + \vec{CB} \Rightarrow \vec{u} = \vec{AC} \quad \boxed{c}$$

✓ ⑤. SABCD pyramid;

$$\vec{SA} + \vec{SC} = 2\vec{SO}$$

$$\vec{SB} + \vec{SD} = 2\vec{SO}$$

$$\Rightarrow \vec{SA} + \vec{SB} + \vec{SC} + \vec{SD} = 4\vec{SO} \Rightarrow \boxed{c=4}$$



✓ ⑥. $B(i, j, k)$; $a \in \mathbb{R}$.

$$u = i + j, v = j + ak, w = i + k.$$

$a = ?$ s.t. (u, v, w) has same or. as B .

$$u = \begin{bmatrix} 1 \\ 1 \\ 0 \end{bmatrix}, v = \begin{bmatrix} 0 \\ 1 \\ a \end{bmatrix}, w = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\Rightarrow K = \begin{bmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & a & 1 \end{bmatrix}$$

$\downarrow \quad \downarrow \quad \downarrow$
 $u \quad v \quad w$

$$\det K = \begin{vmatrix} 1 & 0 & 1 \\ 1 & 1 & 0 \\ 0 & a & 1 \end{vmatrix} = \begin{vmatrix} 1 & 0 & 1 \\ 0 & 1 & -1 \\ 0 & a & 1 \end{vmatrix} = 1 \cdot \begin{vmatrix} 1 & -1 \\ a & 1 \end{vmatrix} =$$

$$= 1 + a > 0 \Rightarrow \boxed{a > -1.}$$

If $\underline{a = -1} \Rightarrow \det K = 0 \Rightarrow \text{lin. indep.} \Rightarrow \text{no basis.}$

✓ ⑦. $K = (O, B)$ right-oriented frame.

$A(1, 2, 0), B(1, 2, 1), C(0, 3, 2), D(1, 1, 1)$. (tetrah.)

$K': (B, \vec{AB}, \vec{AC}, \vec{AD})$.

$$\boxed{A.} \quad [A]_{K'} = M_{B'B} \cdot ([A]_K - [O]_K)$$

$$\vec{BP} = x \vec{AB} + y \vec{AC} + z \vec{AD}$$

$$\vec{BA} = -\vec{AB} \Rightarrow x = -1$$

$$\Rightarrow y = 0$$

$$\Rightarrow z = 0$$

$$\Rightarrow [A]_{K'} = (-1, 0, 0)$$

$$\boxed{B.} \quad \vec{AB} = B - A = (0, 0, 1)$$

$$\vec{AC} = C - A = (-1, 1, 2)$$

$$\vec{AD} = D - A = (0, -1, 1)$$

$$M = \begin{bmatrix} 0 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 2 & 1 \end{bmatrix}$$

$\uparrow \quad \uparrow \quad \uparrow$
 $\vec{AB} \quad \vec{AC} \quad \vec{AD}$

$$\det M = \begin{vmatrix} 0 & -1 & 0 \\ 0 & 1 & -1 \\ 1 & 2 & 1 \end{vmatrix} = 1 \cdot (-1)^4 \cdot \begin{vmatrix} -1 & 0 \\ 1 & -1 \end{vmatrix} = 1 \cdot 1 = 1 > 0$$

$\Rightarrow K, K'$ have the same orientation.

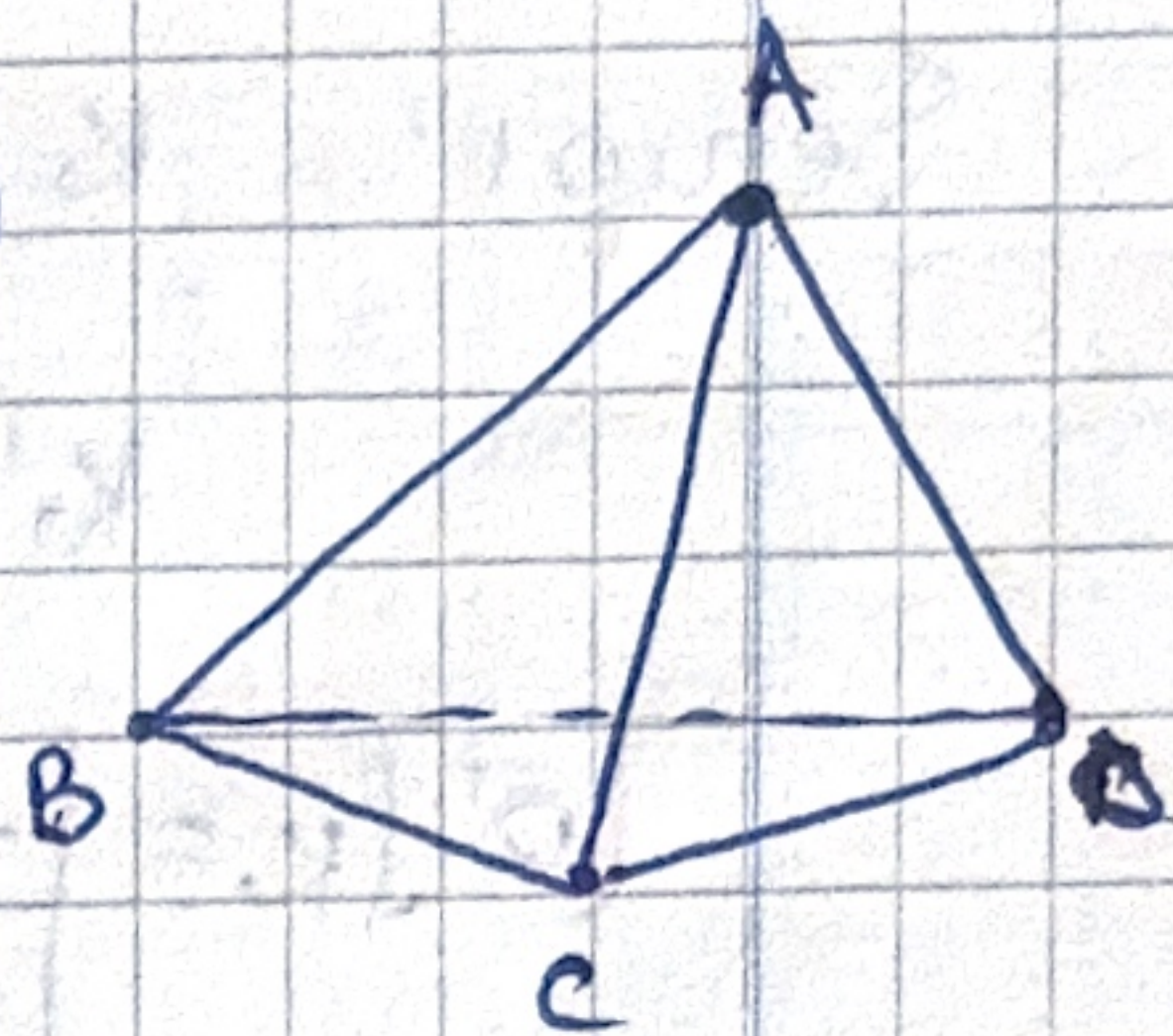
$$\boxed{C.} \quad \vec{AB} = (0, 0, 1)$$

$$\boxed{D.} \quad \vec{BC} = x \vec{AB} + y \vec{AC} + z \vec{AD}$$

$$\vec{BC} = C - B = (-1, 1, 1)$$

$$\vec{BC} = \vec{BA} + \vec{AC} = -\vec{AB} + \vec{AC} \Rightarrow \begin{aligned} x &= -1 \\ y &= 1 \\ z &= 0 \end{aligned}$$

$$\Rightarrow [\vec{BC}]_{B'} = [C]_{B'} = (-1, 1, 0)$$



Setup 1: $K = (O, B)$ frame of $\mathbb{E}^2$.

$K' = (O', B')$ frame with $B' = (i', j')$

$$[O']_K = \begin{bmatrix} -2 \\ -2 \end{bmatrix}, [i']_B = \begin{bmatrix} 3 \\ -1 \end{bmatrix}, [j']_B = \begin{bmatrix} 0 \\ -1 \end{bmatrix}$$

$$[A]_K = \begin{bmatrix} -5 \\ 2 \end{bmatrix}, [B]_K = \begin{bmatrix} -2 \\ 5 \end{bmatrix}, [C]_K = \begin{bmatrix} 1 \\ -1 \end{bmatrix}.$$

✓ (8) $[B]_{K'} = M_{B'B} ([B]_K - [O']_K)$

$$M_{B'B} = \begin{bmatrix} ? \\ ? \end{bmatrix}; M_{BB'} = \begin{bmatrix} 3 & 0 \\ -1 & -1 \end{bmatrix}$$

$$\begin{bmatrix} 3 & 0 & | & 1 & 0 \\ -1 & -1 & | & 0 & 1 \end{bmatrix} \xrightarrow{\frac{1}{3}L_1 + L_2 \rightarrow L_2} \begin{bmatrix} 3 & 0 & | & 1 & 0 \\ 0 & -1 & | & \frac{1}{3} & 1 \end{bmatrix} \quad \begin{matrix} i' \\ j' \end{matrix}$$

$$\xrightarrow{\frac{1}{3}L_1 \rightarrow L_1} \begin{bmatrix} 1 & 0 & | & \frac{1}{3} & 0 \\ 0 & -1 & | & \frac{1}{3} & 1 \end{bmatrix} \xrightarrow{-L_2 \rightarrow L_2} \begin{bmatrix} 1 & 0 & | & \frac{1}{3} & 0 \\ 0 & 1 & | & -\frac{1}{3} & -1 \end{bmatrix}$$

$$[B]_{K'} = \begin{bmatrix} \frac{1}{3} & 0 \\ -\frac{1}{3} & -1 \end{bmatrix} \cdot \left(\begin{bmatrix} -2 \\ 5 \end{bmatrix} - \begin{bmatrix} -2 \\ -2 \end{bmatrix} \right) = \begin{bmatrix} \frac{1}{3} & 0 \\ -\frac{1}{3} & -1 \end{bmatrix} \cdot \begin{bmatrix} 0 \\ 7 \end{bmatrix} = \begin{bmatrix} 0 \\ -7 \end{bmatrix}$$

$$\Rightarrow \boxed{(0, -7)}$$

✓ (9) $\vec{CB} = B - C = \begin{bmatrix} -3 \\ 6 \end{bmatrix}; \vec{CA} = A - C = \begin{bmatrix} -6 \\ 3 \end{bmatrix}$

$$\begin{vmatrix} -3 & -6 \\ 6 & 3 \end{vmatrix} = -9 + 36 = 27 > 0 \Rightarrow \text{same orientation}$$

$\downarrow \quad \downarrow$
 $\vec{CB} \quad \vec{CA}$

$$\boxed{B.} \quad \vec{BA} = A - B = \begin{bmatrix} -3 \\ -3 \end{bmatrix}; \quad \vec{BC} = C - B = \begin{bmatrix} 3 \\ -6 \end{bmatrix}$$

$$\begin{vmatrix} -3 & 3 \\ -3 & -6 \end{vmatrix} = 18 + 9 = 27 > 0 \Rightarrow \text{same or.}$$

$$C. \quad \vec{AB} = B - A = \begin{bmatrix} 3 \\ 3 \end{bmatrix}; \quad \vec{AC} = C - A = \begin{bmatrix} 6 \\ -3 \end{bmatrix}$$

$$\begin{vmatrix} 3 & 6 \\ 3 & -3 \end{vmatrix} = -9 - 18 = -27 \neq 0 \Rightarrow \text{same or.}$$

$$\boxed{D.} \quad \vec{AC} = \begin{bmatrix} 6 \\ -3 \end{bmatrix}, \quad \vec{AB} = \begin{bmatrix} 3 \\ 3 \end{bmatrix}$$

$$\begin{vmatrix} 6 & 3 \\ -3 & 3 \end{vmatrix} = 18 + 9 = 27 > 0. \Rightarrow \text{same or.}$$

Setup 2: $K = (O, B)$ R-or. frame.

$$K' = (O', B')$$

$$[O']_K = \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix}, \quad [i']_B = \begin{bmatrix} 2 \\ 0 \\ -1 \end{bmatrix}, \quad [j']_B = \begin{bmatrix} 0 \\ -1 \\ 0 \end{bmatrix},$$

$$[K']_B = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}, \quad [A]_K, \quad [B]_K, \quad [C]_K, \quad [D]_K$$

✓ (10) $[D]_{K'} = M_{B'B} ([D]_K - [O']_K)$

$$M_{B'B} = \begin{bmatrix} 2 & 0 & 1 \\ 0 & -1 & 0 \\ -1 & 0 & 1 \end{bmatrix} = (-1) \cdot (-1)^4 \cdot \begin{vmatrix} 2 & 1 \\ -1 & 1 \end{vmatrix} = -(2+1) = -3.$$

$$\left[\begin{array}{ccc|ccc} 2 & 0 & 1 & 1 & 0 & 0 \\ 0 & -1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array} \right] \xrightarrow{4-L_3 \rightarrow 4} \left[\begin{array}{ccc|ccc} 1 & 0 & 0 & 1 & 0 & -1 \\ 0 & -1 & 0 & 0 & 1 & 0 \\ 1 & 0 & 1 & 0 & 0 & 1 \end{array} \right]$$

$$L_3 - L_1 \rightarrow L_3$$

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$$-L_2 \rightarrow L_2$$

$$\begin{bmatrix} 1 & 0 & 0 & : & 1 & 0 & -1 \\ 0 & 1 & 0 & : & 0 & -1 & 0 \\ 0 & 0 & 1 & : & -1 & 0 & 2 \end{bmatrix}$$

$$\underbrace{\quad}_{-1} \\ M_{BB'}$$

$$\Rightarrow [D]_{K'} = M_{B'B} \cdot ([D]_K - [O']_K)$$

$$= M_{B'B}^{-1} \cdot ([D]_K - [O']_K)$$

$$= \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 2 \end{bmatrix} \cdot \left(\begin{bmatrix} 2 \\ 1 \\ 0 \end{bmatrix} - \begin{bmatrix} -1 \\ 1 \\ 0 \end{bmatrix} \right)$$

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$$= \begin{bmatrix} 1 & 0 & -1 \\ 0 & -1 & 0 \\ -1 & 0 & 2 \end{bmatrix} \cdot \begin{bmatrix} 3 \\ 0 \\ 0 \end{bmatrix} = \begin{bmatrix} 3 \\ 0 \\ -3 \end{bmatrix}$$

$$(11) \boxed{A} \vec{AB} = B - A = \begin{bmatrix} 0 \\ 2 \\ 0 \end{bmatrix}, \quad \vec{BC} = \begin{bmatrix} -2 \\ 1 \\ 1 \end{bmatrix}, \quad \vec{BD} = \begin{bmatrix} 1 \\ 0 \\ 1 \end{bmatrix}$$

$$\begin{vmatrix} 0 & -2 & 1 \\ 2 & 1 & 0 \\ 0 & 1 & 1 \end{vmatrix} = 2 \cdot (-1)^3 \cdot \begin{vmatrix} -2 & 1 \\ 1 & 1 \end{vmatrix} = -2 \cdot (-2 - 1) = 6 > 0.$$

B.

$\boxed{C}$

D.